GENERAL APTITUDE

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Ratio & Proportion

If A : B = 2 : 3, B : C = 4 : 5 and C : D = 6 : 7. Find A:B:C:D

A. 2 : 3 : 4 : 5　　B. 2 : 12 : 30 : 7　　C. 16 : 24 : 30 : 35　　D. 4 : 5 : 6 : 7

Soln:

A : B = 2 : 3

B : C = 4 : 5

C : D = 6 : 7

A	: B	: C	: D
= ABC	: BBC	: BCC	: BCD
= 2X4X6	: 3X4X6	: 3X5X6	: 3X5X7
= 48	: 72	: 90	: 105
= 16	: 24	: 30	: 35

Ans: C

Ratio & Proportion

Dividing a given number in the given Ratio :

Let T be the Total Amount . Let the given ratio be a:b:c

This means A is divided into three parts such that

First Part = T x a/(a+b+c)

Second Part = T x b/(a+b+c)

Third Part = T x c/(a+b+c)

And First Part + Second Part + Third Part = T

Any Part = Total Amount x (Its related ratio term / Sum of Ratio Terms)

Ratio & Proportion

Q. Find B's share in Rs 6,300 if A:B = 2:3, B:C = 4:5, C:D = 3:7

A.Rs 1080 B. Rs 1800 C. Rs 810 D. Rs 1200

Soln:

A/B	B/C	C/D
2/3	4/5	3/7

A : B = 2 : 3

B : C = 4 : 5

C : D = 3 : 7

A : B : C : D

8 : 12 : 15 : 35

So B's share = 6300 x 12/70 = 1080

Ans : A

Q. Which of the following is a ratio between a number and the number obtained by adding one-fifth of that number to it?

A. 4:5

B. 5:4

C. 5:6

D. 6:5

Ans: C

Ratio & Proportion

Q. Two numbers are in the ratio 7 : 11. If their HCF is 28, then sum of the two numbers is:

A. 112 B. 308 C. 504 D. 196

Soln:

Ans: C

- Let the numbers be 7x and 11x
- HCF of 7x and 11x is x
- HCF = x = 28
- The numbers will be 7 × 28 and 11 × 28
- The numbers will be 196 and 308
- Sum of numbers = 196 + 308
- Sum of numbers = 504
- Sum of numbers is 504

- Let the numbers be 7x and 11x.
- 7x + 11x =18x
- the final number must be the multiple of 18,
- Going by options only 504 is multiple of 18.
- The sum of two number is 504.

Ratio & Proportion

Q. The HCF and LCM of two numbers are 24 and 168 and the numbers are in the ratio 1 : 7. Find the greater of the two numbers.

A. 168 B. 144 C. 108 D. 72

Ans: A

Soln:

- Product of numbers = LCM × HCF
- Let numbers be x and 7x.
- $x \times 7x = 24 \times 168$
- $x^2 = 24 \times 24$
- $x = 24$
- greatest number = $7x = 24 \times 7 = 168$.

Ratio & Proportion(Assignment)

Q. If A:B =2:3, B:C= 4:5 and C:D =6:7 Find A:D is equal to:

A. 16 : 35 B. 8 : 25 C. 4 : 15 D. 2 : 10

Ans : A

Ratio & Proportion(Assignment)

Q. The difference between two positive numbers is 10 and the ratio between them is 5 : 3. Find the product of the two numbers.

A.375 B.175 C.275 D.125 E.250

Ans : A

Ratio & Proportion(Assignment)

Q. Two numbers are in ratio 4 : 5 and their LCM is 180. The smaller number is

A.9 B.15 C.36 D.45

Ans : C

Ratio & Proportion(Assignment)

Q. The average income of all employees is Rs. 20000. The average salary of male employees is Rs. 22000. The average salary of female employees is Rs. 15000. What is the ratio of male employees to female employees?

A. 2 : 5 B. 3 : 4 C. 5 : 2 D. 3 : 5

Ans: C

Ratio & Proportion(Assignment)

Q. The sum of 3 numbers is 98. If ratio between first and second numbers be 2 : 3 and between second and third be 5 : 8, then the second number is?

A. 30 B. 40 C. 50 D. 60

Ans: A

Ratio & Proportion(Assignment)

Two numbers are in ratio 7 : 11. If 7 is added to each of the numbers, the ratio becomes 2 : 3. The smaller number is?

A. 39 B. 49 C. 66 D. 77

Ans: B

Let the numbers be 7x and 11x.

(7x+7)/(11x+7)=2/3

22x+14=21x+21

x=7

Smaller number= 7x = 7 x 7 = 49

Ratio & Proportion(Assignment)

Q. What must be added to each of the numbers 7, 11 and 19, so that the resulting numbers may be in continued proportion?

A. -3 B. -4 C. 3 D. 4

Ans: A

Ratio & Proportion(Assignment)

Q. The incomes of A & B are in the ratio 3:2. Their respective expenditures are in the ratio 5:3. If each of them saves Rs. 2000, what is the income of B?

A. Rs 12,000 B. Rs 8,000 C. Rs 16,000 D. Rs 6,000

Ans : B

Ratio & Proportion(Assignment)

Q. When a particular number is subtracted from each of 7, 9, 11 and 15, the resulting numbers are in proportion. The number to be subtracted is -

A. 1 B. 2 C. 3 D. 5

Ans: C

Sol:

- Let the required number be x
- $\frac{7-x}{9-x} = \frac{11-x}{15-x}$
- (7 - x)(15 - x) = (11 - x)(9 - x)
- 105 - 22x + x2 = 99 - 20x + x2
- 2x = 6
- x = 3

Ratio & Proportion(Assignment)

Q. Average age of three boys is 22 years. If the ratio of their ages is 6 : 9 : 7, then the age of the youngest boy is-

A. 1.8 years B. 9 years C. 18 years D. 16 years

Ans: C

Percentage

- Percentage is a fraction whose denominator is 100(per 100)

Fraction x100	% ÷100	Fraction	%
3/4	75%	5/4	125%
4/5	80%	3/2	150%
2/3	66.66%	1/16	6.25%
5/6	83.33%		
6/5	120%		

Fraction	%	Fraction	%	Fraction	%
1/1	100%	1/6	16.66%	1/11	9.09%
1/2	50%	1/7	14.28%	1/12	8.33%
1/3	33.33%	1/8	12.5%	1/13	7.69%
1/4	25%	1/9	11.11%	1/14	7.14%
1/5	20%	1/10	10%	1/15	6.66%

Percentage

Q. x is 83.33% of y. So y is _____% of x

Solution:

x = 83.33y

$x = \frac{5}{6} y$

So, $y = \frac{6}{5} x$

y = 120% (from chart)

Fraction x100	% 100	Fraction	%
3/4	75%	5/4	125%
4/5	80%	3/2	150%
2/3	66.66 %	1/16	6.25%
5/6	83.33 %		
6/5	120%		

Percentage

Q. x is 80% of y. So y is _____% of x

Solution:

x = 80y

$x = \frac{4}{5} y$

So, $y = \frac{5}{4} x$

y = 125%

Percentage

Q. A number x is increased by 20% then the number is decreased by 20%. Find the net % change.

- **Soln** :
- If a number is increased / decreased by x% then there is always a loss of -$(x/10)^2$
- Net % Change = -$(20/10)^2$ = -(400/100) = - 4% (loss)
- **OR**
- Let the number be 100
- 100 ↑ by 20% =120
- So 20% ↓ of 120 = 96

- 100 120 96

-4% = net change

Percentage

Q. A number x is increased by 50% then the number is increased by 20% and again by 10%. Find the net % change

Soln:

- Let the number be 100
- 100↑ by 50% =150
- Again, 150↑ by 20% =30, So 150 + 30 = 180
- 10%↑of 180 = 18, So, 180 + 18 = 198

- 100 150 180 198

98% = net change

Percentage

- **Two Step change of Percentage**

  In first step if number is changed by a% and the result is again changed by b% the net percentage change of original number is given by

  Net % Change in Number = a + b + ab/100 (+ve or –ve)

Percentage

Q. If a number is increased by 12 % & then decreased by 18% then the net % change in number is

Soln:

Net % Change in Number = a + b + ab/100 (+ve or –ve)

% Change = 12 – 18 + (12 x -18)/100

= -6 – 2.16

= - 8.16%

Percentage

- Expenditure = Price x Consumption
- P $\propto \frac{1}{\text{Consumption}}$
- So, for expenditure to remain constant, when one quantity increases the other quantity should decrease proportionally.
- Eg: If the price of a commodity is decreased by 20% and its consumption is increased by 20%, what will be the increase or decrease in expenditure on the commodity?
- Soln:

Net % Change = a + b + ab/100 (+ve or –ve)

% Change = -20 + 20 + (-20 x 20)/100

= 0 -4 = -4%

OR

100 === 20%↓(Decrease in Price) ===> 80 === 20%↑(Increase in Consumption) ===> 96.

Thus, there is a decrement of 4%

Percentage

Q. Two numbers are respectively 40% and 60% more than a third number. The ratio of the two numbers is:

A. 7:8 B. 3 : 5 C. 4 : 5 D. 6 : 7

Soln:-

- Let the third number be 100
- First number = 40% more than 100 = 100 + 40%of 100 = 100+40 = 140
- Second number = 60% more than 100 = 100 + 60%of 100 = 100 + 60 = 160
- Ratio = $\frac{\text{first number}}{\text{second number}} = \frac{140}{160} = \frac{7}{8}$ = 7 : 8

Ans: A

Percentage using x

Q. Two numbers are respectively 40% and 60% more than a third number. The ratio of the two numbers is:

A. 7:8 B. 3 : 5 C. 4 : 5 D. 6 : 7

Soln:-

- Let the third number be x.
- First number = 40% more than x = x + 40%of x $= x + \frac{40}{100}x = \frac{100x+40x}{100} = \frac{140x}{100}$
- Second number = 60% more than x = x + 60%of x $= x + \frac{60}{100}x = \frac{100x+60x}{100} = \frac{160x}{100}$
- Ratio = $\frac{\text{first number}}{\text{second number}} = \frac{\frac{7x}{5}}{\frac{8x}{5}} = \frac{7}{8}$ = 7 : 8

Ans: A

Percentage(Assignment)

Q. If the price of sugar increases by 25%, by what percent will a housewife have to reduce her consumption to leave total expenditure on sugar unchanged?

A. 25% B. 35% C. 20% D. 15%

Ans: C

Percentage(Assignment)

Q. If the radius of a circle is decreased by 50%, find the percentage decrease in its area.

- A. 55% B. 65% C. 75% D. 85%
- **Soln**:
- Area of a circle = πr^2 where r is the radius
  => Area is directly proportional to r^2
- Assume the old radius is = r1=100
- $A_1 = \pi \times 100^2 = 10000\pi$

  Assume the new radius is = r2=50

  $A_2 = \pi \times 50^2 = 2500\pi$

  Decrease in area = $10000\pi - 2500\pi = 7500\pi$

  Percentage decrease in area = $\frac{difference}{old}$ x100 = $\frac{7500\pi}{10000}$ x 100 = 75%
- **Ans : C**

Percentage(Assignment)

Q. 1.14 expressed as a per cent of 1.9 is:

A. 6% B. 10% C. 60% D. 90%

Ans: C

Percentage(Assignment)

Q. A number x is increased by 20% then the number is increased by 10% and again by 50%. Find the net % change.

A. 77%   B. 75%   C. 88%   D. 98%   E. 99%

Ans : D

Percentage(Assignment)

Q. If the altitude of a triangle increases by 5% and the base of the triangle increases by 7%, by what percent will the area of the triangle increase?

A. 12.25% B. 12.35% C. 6.00% D. 5.25%

Ans B

Percentage(Assignment)

Q. The length and breadth of a room are increased by 25% and 40% respectively. While the height is decreased by 20%.Find % change.

A. 16% B. 40% C. 60% D. 30%

Ans B

Percentage(Assignment)

Q. If the length of a rectangle is increased by 37.5% and its breadth is decreased by 20%, find the change in its area.

A. 15% increase B. 13% decrease C. 10% increase D. 10% decrease

Ans: C

Percentage(Assignment)

Q. The ratio 5 : 4 expressed as a percent equals :

A. 125%   B. 80%   C. 40%   D. 12.5%

Ans: A

Required % = 5/4 × 100 = 125%

Percentage(Assignment)

Q. 12% of 5000 = ?

A. 600 B. 620 C. 680 D. 720

Ans: A

Percentage(Assignment)

Q. 280% of 3940 = ?

A. 10132

B. 11032

C. 11230

D. 11320

Ans: B

Percentage(Assignment)

Q. 15% of 578 + 22.5% of 644 = ?

A. 231.4

B. 231.6

C. 231.8

D. 233.6

Ans: B

Calendar

- In Non Leap year –
  - 365 days
  - 1 year = 52 weeks + 1 odd day(extra day)
  - 28^{th} February
- In Leap year –
  - 366 days
  - 1 year = 52 weeks + 2 odd days
  - 29^{th} February
- A **century leap year** is a **year** that is exactly divisible by 400
  - **years** 1600 and 2000 were **century leap years**; (400,800,1200,1600,2000 – century leap years till date)
  - **years** 1700, 1800, and 1900 were not **century leap years**.
- To find the day of a week on a given date we use the concept of "**odd days**".
- 01/01/0001 A.D(Anno Domini) was a Monday and 1^{st} day of week so 1^{st} January 0001 was a Monday.

Calendar

- In a century,
  - 24 leap year
  - 76 non leap years

100 years

Leap year — non leap year

24x2 + 76 x 1

$=\frac{48}{7}$ $=\frac{76}{7}$

remainder

6 + 6 $= 12 \div 7 = 5 \leftarrow$ remainder

5 extra(odd) days in a century (100 years)
100 years = 5 odd days ⟵ remainder
200 years = $10 \div 7 = 3$ odd days
300 years = $15 \div 7 = 1$ odd days
400 years = 0 odd days (as century leap year)

Calendar

Years	No. of odd
Ordinary year	1
Leap year	2
100 years	5
200 years	3
300 years	1
400 years	0

Calendar

Day of week	No. of odd
Sunday	0
Monday	1
Tuesday	2
Wednesday	3
Thursday	4
Friday	5
Saturday	6

Calendar

Month		Remainder
January	31 ÷ 7	3
February	28 ÷7 or 29 ÷ 7	0(non leap) or 1(leap)
March	31 ÷ 7	3
April	30 ÷ 7	2
May	31 ÷ 7	3
June	30 ÷ 7	2
July	31 ÷ 7	3
August	31 ÷ 7	3
September	30 ÷ 7	2
October	31 ÷ 7	3
November	30 ÷ 7	2
December	31 ÷ 7	3

Calendar

Q. What was the day of the week on 15th August, 1947?

Soln:

Completed till 1946

1946

$\frac{1900}{400} = 300$ $\frac{46}{4} = 11$(quotient)

1 odd day 46 + 11 = 57 $\frac{57}{7} = 1$(remainder)

In 1946, odd days are,

1900 46

1 + 1 = 2 odd days

1946 month date

Total odd days = 2 + 2 + 1 = 5 odd days

As per table for days of a week , 5 $\longleftrightarrow$ Friday

As month is August, go till July as per table,

J F M A M J J

3 + 0 + 3 + 2 + 3 + 2 + 3 = 16

Now, $\frac{16}{7} = 2$ (remainder)

For date ,

$\frac{15}{7} = 1$ (remainder)

Calendar

For Months -

J	F	M	A	M	J	J	A	S	O	N	D
0	3	3	6	1	4	6	2	5	0	3	5

For years -

1600 – 1699	6
1700 – 1799	4
1800 – 1899	2
1900 – 1999	0
2000 – 2099	6

Calendar

Q. What day of the week was 29 June 2010 ?

- **Soln**

1. Last 2 digits of the year → 10
2. Divide by 4 (10 ÷ 4) = 2(quotient)
3. Take the date → 29
4. Take the no. of month → 4(from table)
5. Take the no. of year → 6 (from table)

51 (add)

6. Divide by 7 → $\frac{51}{7}$ = 2(remainder)

Check table for day of the week

2 ⟷ Tuesday

Calendar

Q. What was the day of the week on 29th February, 2012?

Soln:

1. Last 2 digits of the year → 12
2. Divide by 4 (12 ÷ 4) = 03(quotient)
3. Take the date → 29
4. Take the no. of month → 03 (from table)
5. Take the no. of year → 06 (from table)

53 (add)

6. Divide by 7 → $\frac{53}{7}$ = 4 (remainder)

subtract 1 from remainder

In this case for all dates of **January & February** in a leap year , 4 -1 =3

Check table for day of the week

3 ⟷ Wednesday

Calendar

It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010?

A. Sunday B. Saturday C. Friday D. Wednesday

Ans: C

On 31st December, 2005 it was Saturday.

Number of odd days from the year 2006 to the year 2009 = (1 + 1 + 2 + 1) = 5 days.

On 31st December 2009, it was Thursday.

on 1st Jan, 2010 it is Friday.

Calendar

Q. If we have preserved the calendar of 2017. Find the next immediate year in which we can reuse.

A. 2027 B.2023 C. 2025 D. 2029

Soln:

x/4 (x = given year)

$\frac{2017}{4}$ = 1 (remainder)

For any year divide by 4, the possibility of remainder is 0,1,2,3

If remainder = 0 → x + 28

If remainder = 1 → x + 6

If remainder = 2/3 → x + 11

So, $\frac{2017}{4}$ = 1(remainder)

2017 + 6 = 2023 **Ans: B**

Calendar

Q. Which of the following days can never be the last day of a century?

A. Sunday B. Monday C. Tuesday D. Wednesday

- **Soln**:
- The last day of century can be only
- 1 odd day(Monday)
- 3 odd days (Wednesday)
- 5 odd days (Friday)
- 7 or 0 odd days (Sunday)
- So, century can never end in **Tuesday** , **Thursday** or **Saturday**.
- **Ans: C**

Calendar(Assignment)

Q. What was the day of the week on 26th January, 1947?

Soln:

1. Last 2 digits of the year → 47
2. Divide by 4 (47 ÷ 4) = 11(quotient)
3. Take the date → 26
4. Take the no. of month → 0 (from table)
5. Take the no. of year → 0 (from table)

84 (add)

6. Divide by 7 → $\frac{84}{7}$ = 0 (remainder)

Check table for day of the week

0 ⟷ Sunday

Calendar(Assignment)

- Q. The day on 5th April of a year will be the same day on 5th of which month of the same year?
- A. 5th July B. 5th August C. 5th June D. 5th October

- **Ans A**

- April & July for all years have the same calendar. So, a day on any date of April will be the same day on the corresponding date in July.
- The same day will fall on 5th July of the same year.

Calendar(Assignment)

Q. What was the day of the week on your birthdate?

Q. 13th October 2019 is a Sunday. Find the day on 13th October 1989?

A. Sunday B. Monday C. Friday D. Wednesday

Ans: C

Q. 1st March 2006 falls on a Wednesday .What day does 1st March 2010 fall on?

A. Tuesday B. Monday C. Friday D. Wednesday

Ans: B

Q. Today is Monday. Which day will be after 64 days?

A. Tuesday B. Monday C. Friday D. Wednesday

Ans: A

Q. Today is Monday. After 30 days it will be?

A. Tuesday B. Monday C. Friday D. Wednesday

B. Ans: D

Calendar(Assignment)

Q. 15th August 1947 was a Friday. Find the day on 15th August 1977?

- Soln:

$$\begin{array}{r} 1977 \\ -\ 1947 \\ \hline 30 \text{ years} \end{array}$$

Leap years between 1947 to 1977

1948	1964	8 years
1952	1968	
1956	1972	
1960	1976	

30 + 8 = 38

total years leap

$\frac{38}{7}$ = 3 (remainder)

As 15th August 1947 was a Friday ,

So, Friday + 3 days = **Monday**

Calendar(Assignment)

Q. 4th January 2016 falls on Monday. What day of the week does 4th January 2017 lies?

A. Wednesday B. Thursday C. Tuesday D. Monday

Soln:

Normal year = 1 odd day

Leap year = 2 odd days

Jan 4, 2016 → Monday

\+ 2 (as leap year)

Jan 4,2017 → Wednesday

Ans: A

Calendar(Assignment)

Q. Wednesday falls on 5th of a month .So which day will fall 5 days after 22^{nd} of the same month?

A. Tuesday B. Friday C. Thursday D. Wednesday

Ans: B

5^{th} = Wednesday

+7

12^{th} = Wednesday

+7

19^{th} = Wednesday

22^{nd} = Saturday

+5

27^{th} = Thursday

5 days after 22^{nd} will be **Friday**

Calendar(Assignment)

Q. What dates of May 2002 did Monday fall on?

Soln:

Lets take date = 1st May 2002

1. Last 2 digits of the year → 02
2. Divide by 4 (02 ÷ 4) = 00(quotient)
3. Take the date → 01
4. Take the no. of month → 01 (from table)
5. Take the no. of year → 06 (from table)

10 (add)

6. Divide by 7 → $\frac{10}{7}$ = 3 (remainder)

Check table for day of the week

3 ⟷ Wednesday

1st May 2002 falls on Wednesday

1	2	3	4	5	6
W	Th	F	Sa	Su	M

first Monday

Now add 7 to it to find remaining Mondays

Dates on which Monday falls are -
6 , 13 , 20, 27

Calendar(Assignment)

Q. On what dates of April, 2001 did Wednesday fall?

A. 1st, 8th, 15th, 22nd, 29th B. 2nd, 9th, 16th, 23rd, 30th C. 3rd, 10th, 17th, 24th

D. 4th, 11th, 18th, 25th

Ans: D

Calendar(Assignment)

Q. What is the day on 22 April 2222?

A. Monday B. Tuesday C. Saturday D. Sunday

Ans: A

Calendar(Assignment)

Which of the following is not a leap year?

A. 700 B. 800 C. 1200 D. 2000

Ans: A

The century divisible by 400 is a leap year.
The year 700 is not a leap year.

Calendar(Assignment)

Q. Today is Monday. Which day will be on 61st day?

Soln:

1 week = 7 days. Taking the multiple of 7

56 - Monday	or	63 - Monday
57 – Tuesday		62 - Sunday
58 – Wednesday		61 - Saturday
59 – Thursday		
60 – Friday		
61 - Saturday		
56 + 5 = 61 days		63 – 61 = 2 days
(add 5 days)	or	(subtract 2 days)

Calendar(Assignment)

Q. January 1, 2007 was Monday. What day of the week lies on Jan. 1, 2008?

A. Monday

B. Tuesday

C. Wednesday

D. Sunday

Ans: B

Mixtures & Alligation

- **Alligation** : It is the rule which enables us to find the ratio in which two or more ingredients at given prices must be mixed to produce a mixture of a desired price.(mixing / linking)
- **Mean Price** : The cost price of a unit quantity of mixture is called the mean price.
- **Dearer** : The more expensive ingredient

- Note :

Always maintain the order in which problem is given else answer gets changed

Mixtures & Alligation

Type 1 oranges at Rs.60 per kg and Type 2 oranges at Rs.120 per kg and when mixed cost is Rs.75 per kg. Find the ratio in which Type 1 and Type 2 oranges are mixed.

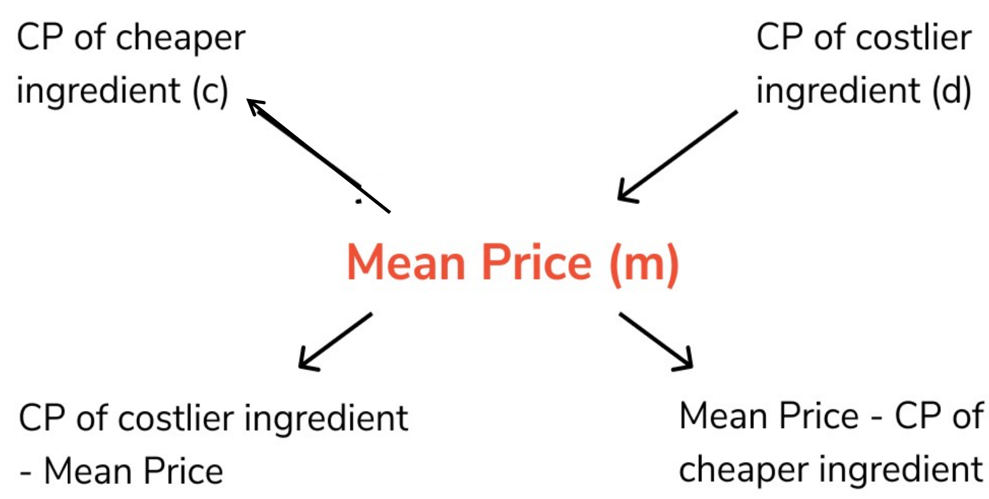
Soln:

Type 1 60 | Type 2 120

75

x = d-m | y =m-c

$$\frac{x}{y} = \frac{d-m}{m-c} = \frac{120-75}{75-60} = \frac{45}{15} = \frac{3}{1} = 3:1$$



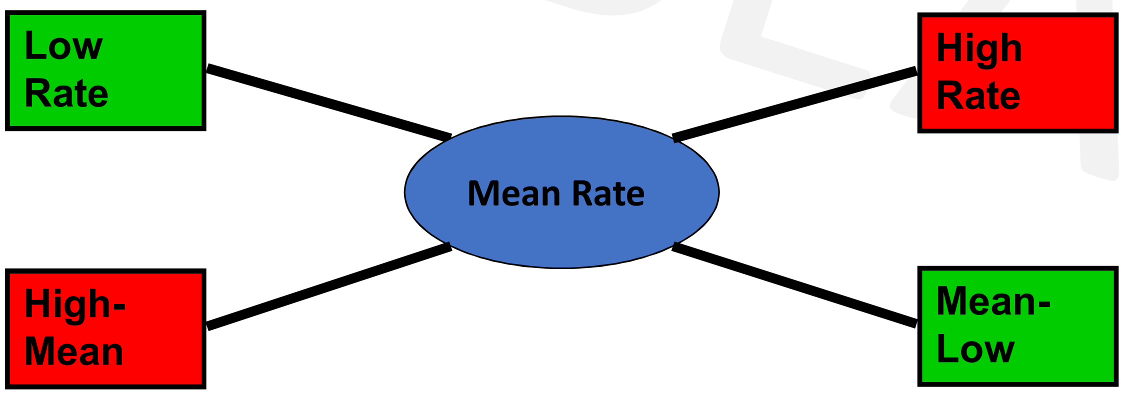
$$\frac{\text{Quantity of cheaper ingredient}}{\text{Quantity of costlier ingredient}} = \frac{d - m}{m - c}$$

Mixtures & Alligation

$$\frac{\text{Quantity of Lower}}{\text{Quantity of Higher}} = \frac{(\text{C.P. of Higher}) - (\text{Mean Price})}{(\text{Mean Price}) - (\text{C.P. of Lower})}$$

$$\frac{\mathbf{Q}l}{\mathbf{Q}h} = \frac{\mathbf{CP}h - \mathbf{CP}m}{\mathbf{CP}m - \mathbf{CP}l}$$

(**Qty** Low) : (**Qty** High) = (**CP**h-**CP**m) : (**CP**m-**CP**l)



THANK YOU!